Certificate

Name: HARSH RAJ SHARMA Class: 1st Sem

Roll No: AB-16030 Exam No:

Institution S.G.S.I.T.S., Indore

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Group - B

* **Aim:-** Determination of total hardness by EDTA method

* **Apparatus:-** Conical flask, pipette, burette etc

* **Solution required:-** Known water sample, unknown water sample, Eriochrome Black T - indicator, EDTA solution, Buffer solution at pH 10.

* **Theory:-** Hardness of water can be determined by complexometric titration. Ethylene - diaminetetraacetic acid (EDTA) is a well known complexing agent, which is widely used in analytical work on account of its powerful complexing action & commercial availability. The EDTA method is based upon the fact that ethylene - diaminetetraacetic acid (EDTA) forms remarkably stable complex compound with all metals particularly with di & polyvalent metals.

Function of EDTA:- For all practical purposes, the hydrated disodium salt of the reagent is used which has the following formula:- $C_{10}H_{16}N_2O_8Na_2$

The formula may be abbreviated for the sake of simplicity as Na_2H_2Y

EDTA ion reacts with calcium & magnesium forming stable complexes, the calcium - EDTA complex is more stable than magnesium - EDTA complex which is more stable than magnesium indicator complex.

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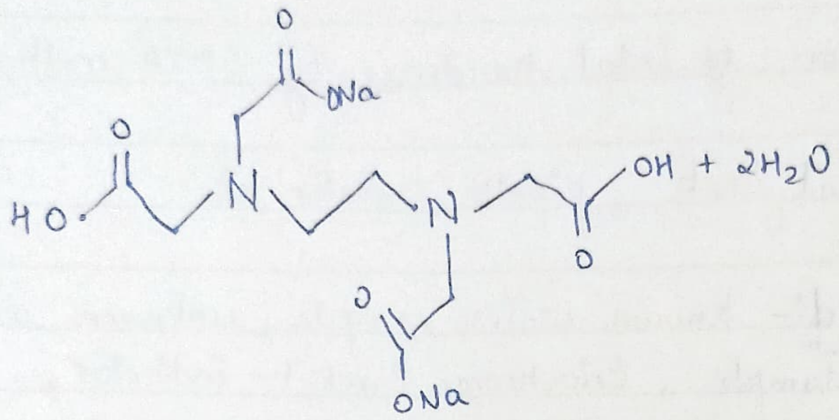


<u>Chemical structure of disodium salt of EDTA</u>

- $Na_2H_2Y \xrightarrow{pH\ 10} 2Na + H_2Y^{2-}$

- $H_2Y^{2-} + Ca^{2+} \xrightarrow{pH\ 10} CaH_2Y$ (Ca - EDTA complex)

- $H_2Y^{2-} + Mg^{2+} \xrightarrow{pH\ 10} MgH_2Y$ (Mg - EDTA complex)

The relative stability of the complenes so far considered is as follows:-

Ca - EDTA complen > Mg - EDTA complen > Mg - Indicator complen

Function of indicator:- The indicator Eriochrom black - T is a tribasic dye, the abbreviated formula of which is written as H_3In & which dissociates producing different colored ions at various pH as follows:-

$$H_3In \rightleftharpoons H_2In^- \rightleftharpoons HIn^{2-} \rightleftharpoons In^{3-}$$

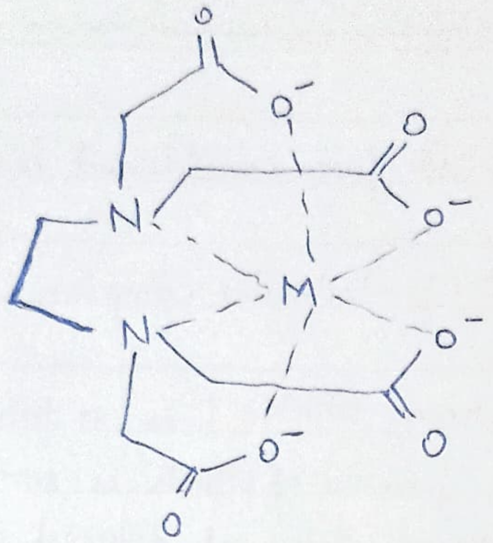
H_2In^-	HIn^{2-}	In^{3-}
Red (below pH 6)	Blue (pH 7-11)	Orange (above pH 12)

The bivalent blue colored indicator ion which existed about pH 10 react with magnesium producing red coloured magnesium indicator complen.

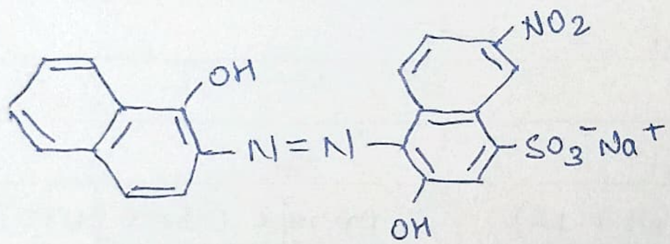
$$HIn^- + Mg^{++} \longrightarrow MgIn^- + H^+$$

HIn^-	Mg^{++}	$MgIn^-$
Free indicator (Blue)	Analyte (colorless)	Complen (Wine red)

So, if a drop of indicator is added to hard water sample containing Ca^{2+} & Mg^{2+} ions at pH 10, the magnesium indicator complen will be formed producing wine red colour.



Chemical structure of Metal - EDTA complex



Chemical structure of eriochrome black - T indicator

- $HIn^- + Mg^{++} \longrightarrow MgIn^- + H^+$

  Free indicator (Blue) + Analyte (Colorless) → Complex (wine red)

- $MgHIn + H_2Y^{2-} \longrightarrow MgH_2Y + HIn^{2-}$

  wine red → Blue

Now, to this solution, if EDTA is gradually added, it will remove calcium ion first because Ca - EDTA complex is more stable. When all Ca^{2+} ions are used up, excess EDTA, if added will snatch Mg^{2+} ions from Mg - indicator complex because Mg - EDTA complex is more stable than Mg - indicator complex leaving bivalent indicator ion free, which being blue in colour will produce blue coloured solution. Thus, a change from red colour to blue colour will indicate the end point.

$$\underset{\text{Wine red}}{MgHIn} + H_2Y^{2-} \longrightarrow MgH_2Y + \underset{\text{Blue}}{HIn^{2-}}$$

* **Procedure:-**

1.) Pipette out 20 ml of the known water sample into a 100 ml conical flask.
2.) Add 2 ml of the buffer solution & 2 drops of Eriochrome black - T indicator. The solution becomes wine red.
3.) Titrate it with EDTA solution from the burette until color changes to clear blue at the end point. Note the burette reading.
4.) Repeat the same process, till you get two concurrent readings.
5.) Now titrate the unknown water sample in the similar manner till two concardant readings are obtained.

Observation for known solution :-

S.NO.	Volume of Known Solution (ml).	Volume of Intermediate Solution (ml)		Total Volume of Intermediate (ml)
		Initial Reading	Final Reading	
1.	20 ml	0 ml	17.5 ml	
2.	20 ml	0 ml	17.2 ml	17.2 ml
3	20 ml	0 ml	17.2 ml	

Observation for Unknown Solution :-

S NO.	Volume of Known Solution (ml)	Volume of Intermediate Solution (ml)		Total Volume Intermediate (ml)
		Initial Reading	Final Reading	
1.	20 ml	0 ml	8.3 ml	
2.	20 ml	0 ml	8.3 ml	8.3 ml.
3.	20 ml	0 ml	8.3 ml	

* Calculations :-

$N_1V_1 = N_2V_2$

$N_2 = \frac{N_1V_1}{V_2}$

$N_2V_2' = N_3V_3$

$N_3 = \frac{N_2V_2'}{V_3}$

$N_1 \Rightarrow$ Normality of known solution
$V_1 \Rightarrow$ Volume of known solution
$N_2 \Rightarrow$ Normality of intermediate solution
$V_2 \Rightarrow$ Volume of intermediate solution
$N_3 \Rightarrow$ Normality of alloy solution
$V_3 \Rightarrow$ Volume of alloy solution
$V_2' \Rightarrow$ Volume of intermediate solution with unknown solution

$N_1 = \frac{1}{40} \Rightarrow V_1 = 20 ml \quad V_2 = 17.2 ml$

$N_2 = \frac{1}{40}\left(\frac{20}{17.2}\right) = \frac{1}{34.4} \Rightarrow N_2V_2 = N_3V_3 \Rightarrow \frac{1}{34.4} \times 3.3 = N_3 \times 20$

$\therefore N_3 = \frac{3.3}{(34.4)(20)}$

After knowing the normality of unknown solution (N_3), we can convert it into strength by multiplying it with equivalent weight of $CaCO_3$, Equivalent weight of $CaCO_3 = 50$.

$N_3 \times$ Equivalent weight of $CaCO_3$ = Hardness of given water sample OR Strength in g/L of $CaCO_3$

$\Rightarrow \frac{3.3}{34.4 \times 20} \times \frac{100}{2}$

Strength $\times 10^3$ = Hardness of given water sample in ppm
$= 8.3/34.4 \times 2 \times \frac{100}{2} \times 10^3 = 603.2$ ppm

*** Result :-** The EDTA hardness of the given water sample is 603.2 ppm.

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★ **Aim** :- Determination of the carbonate, bicarbonate & total alkalinity of given water sample by neutralization titration.

★ **Apparatus** :- Conical flask, pipette, burette etc.

★ **Solution required** :- N/30 Sodium carbonate solution, hydrochloric acid solution, phenolphthalein and methyl orange indicator.

★ **Theory** :- The alkalinity of natural water is generally due to the presence of carbonates & bicarbonates of calcium and magnesium. In industries, hard water is softened by lime soda process due to which the resulting water contains considerable amount of CO_3^- & OH^- which leads to alkalinity. Highly alkaline water may lead to caustic embrittlement & also may cause deposition of precipitates & sludge in boiler tubes & pipes. Depending on the anions that are present in water, alkalinity is classified as :-

1.) Hydroxides only
2.) Carbonates only
3.) Bicarbonates only
4.) Hydroxides & Carbonates
5.) Carbonates & Bicarbonates

The possibility of hydroxides & bicarbonates existing together is ruled out because of the fact that they combine with each other as follow forming carbonates.

$$OH^- + HCO_3^- \longrightarrow CO_3^{2-} + H_2O$$

- $OH^- + HCO_3^- \longrightarrow CO_3^{2-} + H_2O$

  hydroxide combines with bicarbonate

- $OH^- + H^+ \longrightarrow H_2O$
- $CO_3^{2-} + H^+ \longrightarrow HCO_3^-$
- $HCO_3^- + H^+ \longrightarrow H_2O + CO_2$

The type & extent of alkalinity present in a water sample may be conveniently determined by titrating aliquot of the sample with a standard acid to phenolphthalein end-point, p and continuing the titration to the methy orange end-point, m.

The volume of acid run-down up to phenolphthalein end-point (p) corresponds to the complete neutralization of OH^- ions & half neutralization of CO_3^{2-} ions, while the volume of acid-run after p corresponds to the complete neutralization of HCO_3^-. The total amount of acid used from beginning of the experiment corresponds to the total alkalinity present, i.e., those formed from half neutralization of CO_3^{2-} & those already present in the water sample.

* Procedure :-

1.) Pipette out 20ml of the known solution into the conical flask.
2.) Now add 2 drops of methyl orange indicator. Titrate the solution against acid solution till color changes from yellow to pink.
3.) Note the volume of acid used.
4.) Repeat the same experiment till two concordant readings are obtained.
5.) Now pipette out 20ml of unknown water sample into the conical flask, Add two drops of phenolphthalein indicator. The solution becomes pink indicating presence of CO_3^{2-} ions.
6.) Titrate the pink solution against acid till solution becomes colorless. Note down the reading of the burette

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Observations of known solution :-

S.NO.	Volume of Known solution (ml)	Volume of Intermediate solution (ml)		Total Volume of Intermediate sol^n (ml)
		Initial Reading	Final Reading	
1.	20 ml	0 ml	18.8 ml	
2.	20 ml	0 ml	16.3 ml	16.3 ml
3.	20 ml	0 ml	16.3 ml	

Observation of unknown solution :-

S NO.	Volume of known solution (ml)	Volume of Intermediate solution (ml)		Total Volume of Intermediate sol^n (ml)
		Initial Reading	Final Reading	
1.	20 ml	0 ml	13 ml	
2.	20 ml	0 ml	12.1 ml	12.1 ml
3.	20 ml	0 ml	12.1 ml	

7.) At this stage, add two drops of methyl orange indicator to the same solution. Titrate the solution against acid till solution becomes pink. Note the burette reading.

8.) Repeat the experiment till two concordant readings are obtained.

★ Calculations :-

given = $N/30$ (Known solution)

N_1 = Normality of known solution

V_1 = Volume of known solution

N_2 = Normality of intermediate solution

V_2 = Volume of intermediate solution

$N_1V_1 = N_2V_2$

$1/30 \times 20 = N_2 \times 16.3$

$\therefore N_2 = 0.04089$

★ Phenolphthalein titration :-

N_2 = Normality of intermediate solution

V_3 = Volume of unknown solution

N_p = Normality of unknown solution (p)

p = Volume of intermediate solution with phenolphthalein indicator

$$N_2 \times p = N_p \times V_3$$

$$\therefore N_p = \frac{N_2 \times p}{V_3}$$

$$0.04089 \times 4 = N_p \times 20$$

Hence, $N_p = 0.008178$

★ Methyl Orange Titration:-

N_2 = Normality of intermediate solution
V_3 = Volume of unknown solution
N_m = Normality of unknown solution (m)
m = Volume of intermediate solution with methyl orange indicator

$$N_2 \times m = N_m \times V_3$$

$$N_m = \frac{N_2 \times m}{V_3}$$

$$\therefore 0.04089 \times 12.1 = N_m \times 20$$

Hence, $N_m = 0.02473845$

After knowing the normality of unknown solution (N_p, N_m), we can convert it into strength by multiplying it with equivalent of $CaCO_3$

Phenolphthalein = $N_p \times$ eq. weight of $CaCO_3 \times 10^3$
$= 0.008178 \times 50 \times 10^3$
$= 408.9$ ppm

Methyl orange = $N_m \times$ eq. weight of $CaCO_3 \times 10^3$

$= 0.024738 \times 50 \times 10^3$

$= 1236.9$ ppm

★ Results: -

Total alkalinity (T) = P + M

$= (1236.9 + 408.9)$ ppm

$= 1645.8$ ppm

Carbonate alkalinity = 2P

$= 2 \times 408.9$ ppm

$= 817.8$ ppm

Bicarbonate alkalinity = T − 2P

$= (1645.8 - 817.8)$ ppm

$= 828$ ppm

★ Precautions: -

1.) Stir the mixture with a clean glass rod.

2.) Using a wash bottle wash all traces of the solute from the clock glass into the beaker.

3.) Fill the burette untill the bottom of the meniscus is on the mark.

★ **Aim:-** To determine percentage purity of Iron alloy by internal indicator method.

★ **Apparatus:-** Conical flask, pipette, burette etc.

★ **Solutions required:-** N/40 ferrous sulphate solution, diphenylamine as an internal indicator, potassium dichromate solution, a mixture of H_2SO_4 & H_3PO_4, alloy solution (Concentration 4g/L)

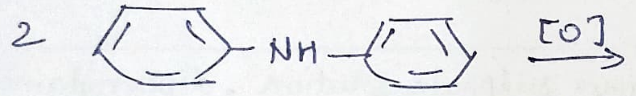
★ **Theory:-** $K_2Cr_2O_7$ contains oxygen available for oxidation which it loses in presence of a substance capable for receiving it, & in presence of an excess of sulphuric or hydrochloric acid. Thus, a ferrous salt in an acidified solution is readily oxidized to ferric salt by the addition of a solution of potassium dichromate.

At the beginning of titration, there are only ferrous ions, as the titration proceeds these ions are converted into ferric ions. At the end point, all the ferrous ions are converted into ferric ions. Gram equivalent weight of the $K_2Cr_2O_7$ used upto the end point will correspond to the gram equivalent of ferrous ions present in the solution of alloy.

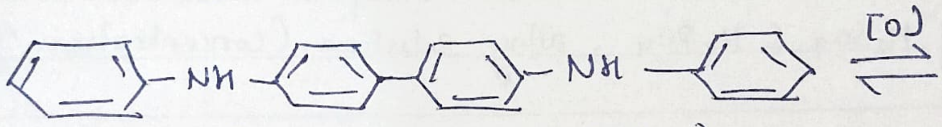
The end point of the reaction can be detected using diphenylamine as an indicator, 1% solution of diphenylamine in concentrated sulphuric acid is used for the titration of iron (II) with potassium dichromate solution.

- $FeSO_4(NH_4)_2SO_4 \cdot 6H_2O \longrightarrow FeSO_4 + (NH_4)_2SO_4 + 6H_2O$

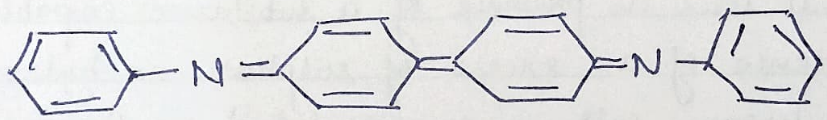
- $6FeSO_4 + K_2Cr_2O_7 + 7H_2SO_4 \longrightarrow 3Fe_2(SO_4)_3 + Cr_2(SO_4)_3 + K_2SO_4 + 7H_2O$

- 

(I) diphenylbenzidine (colorless)



(II) Diphenylamine (colorless)



(III) Diphenylbenzidine (violet)

Diphenylamine is first ~~as~~ oxidized ~~in~~ to biphenyl benzidine (colorless) which is the real indicator. It is reversibly oxidized to its quinonoid form (violet color). As it shows different colors in reduced & oxidized form, it acts as a redox indicator.

Red-Ox potential of In $(0.5-1\,M\ H_2SO_4 + 0.5\,M\ H_3PO_4)$ medium, :-

$$E^{o'}_{In} = -0.76V$$

Suitable for titration of Fe^{2+} by dichromate solution, in presence of H_3PO_4, :-

$$E^{o'}_{In} = -0.61\ V$$

During the titration, oxidation of two species takes place. The first one is the oxidation of ferrous into ferric ions & another one is the oxidation of diphenylamine into diphenylbenzidine.

The oxidation potential of the two system is given by Nernst's equation as follows :-

$$E = E^{o} + \frac{RT}{nf} \ln \frac{[\text{Oxid.}]}{[\text{Red.}]}$$

And, the formal potential of system is :-

$Fe^{2+} \rightarrow Fe^{3+}$ is $0.68\ V$.

Similarly, formal potential of indicator system is $0.76V$. The system having low oxidization potential will oxidize first. Due to oxidation of Fe^{2+} into Fe^{3+}, concentration of Fe^{3+} increases. It results in the increase in the potential of the system. If the potential of the system exceeds $0.76\ V$, it results in the oxidation

- Formal potential of system $Fe^{2+} \rightarrow Fe^{3+}$ is 0.68 V

- $H_3PO_4 + Fe^{3+} \longrightarrow (FeHPO_4)^+ + 2H^+$

  Green colored complex

of indicator even before the actual end point. So, phosphoric acid is added to reduce the formal potential of ferrous-ferric ion system.

Excess of Fe^{3+} ions are removed by phosphoric acid which forms stable green coloured complex with Fe^{3+}

Due to formation of this complex, the potential of the system remain below 0.76 V till the end point. At end point, one drop of $K_2Cr_2O_7$ will oxidize the indicator & color of the solution changes to blue violet.

★ Procedure :-

1.) Pipette out 20 mL of the known solution into the conical flask. Add about 2 mL of acid mixture & 2 drops of diphenylamine indicator into it.

2.) Now, titrate it against $K_2Cr_2O_7$ solution till solution becomes blue. Note down the reading & repeat the experiment for two concordant observations.

3.) Similarly, repeat the experiment with unknown alloy solution.

★ Calculation :-

N_1 = Normality of known solution
V_1 = Volume of known solution
N_2 = Normality of intermediate solution
V_2 = Volume of intermediate solution

N_3 = Normality of alloy sol^n
V_3 = Volume of alloy sol^n
V_2' = Volume of intermediate sol^n with alloy sol^n

Observation for known solution :-

S.NO.	Volume of known solution (ml)	Volume of $K_2Cr_2O_7$ (ml)		Total Volume of $K_2Cr_2O_7$ used
		Initial reading	Final reading	
1.	20 ml	0 ml	18.6 ml	
2.	20 ml	0 ml	18.6 ml	18.6 ml
3.	20 ml.	0 ml.	18.6 ml.	

Observation for unknown solution :-

S.NO.	Volume of Known solution (ml)	Volume of $K_2Cr_2O_7$ (ml)		Total Volume of $K_2Cr_2O_7$ used
		Initial reading	Final reading	
1.	20 ml	0 ml	10.5 ml	
2.	20 ml	0 ml	10.5 ml	10.5 ml
3.	20 ml.	0 ml.	10.5 ml	

$N_1V_1 = N_2V_2$

$\therefore N_2 = \frac{N_1V_1}{V_2} = \frac{1 \times 20 \times 10}{30 \times 156}$

Hence, $N_2 = 0.035$

$N_2V_2' = N_3V_3$

$\therefore N_3 = \frac{N_2V_2'}{V_3} = \frac{30 \times 10^{-3} \times 10^3}{10 \times 20}$

Hence, $N_3 = 0.018$

% of Iron in given alloy solution = Strength in g/L $\times 55.8 \times 100/4$

$= 0.018 \times 55.8 \times 25$

$= 25.11$

* Precautions :-

1.) Make sure to always vigorously shake the flask after every drop if the end point is still unknown.

2.) Bubble present in the nozzle of the burette, must be removed before taking the initial reading.